Supplementary material for "CertiVar: Certified Likelihood Ratios for Variant-Effect Predictor Calls, Calibrated on Saturation Mutagenesis Data"

Aaryav Sharma
Eastview High School, Apple Valley, MN, USA

Section, equation, figure and table numbers without the prefix S refer to the main text. Contents: S1 proofs of Propositions 1–3, Theorems 1–2 and Corollary 1; S2 simulation (Fig. S1) and why marginal thresholds do not control the minority class; S3 calibration size, region and gene transfer (Fig. S2A–C), per-gene coverage (Table S1) and prior estimation (Table S2); S4 strata, assay-label sensitivity (Table S3), calls judged against ClinVar labels (Table S4), ClinVar VUS by assay status and the calls on them (Tables S5–S6) and the oracle comparison (Fig. S3); S5 implementation and the comparison rules; S6 experimental designs (Table S7) and all baselines by gene and predictor (Table S8); S7 coverage and certification figures (Figs. S4–S5) and further results.

S1 Proofs

Assumptions, propositions, theorems and the corollary are stated in Section 3.1; Q, C, $\hat{t}_k$, the call rates λ_k, ϕ_k and their bounds, L^+ and U^- are defined in Section 3.

Throughout, fix a class k and write F_k for the distribution function of the nonconformity r_k of a variant drawn from $Q(\cdot \mid Y = k)$. By Assumption 1, the kept calibration variants (one per residue of $\mathcal{A}$) and the variant to be called are independent draws from Q; conditionally on all labels, the n_k kept variants of class k and a class-k test variant are independent draws from $Q(\cdot \mid Y = k)$.

Proposition 1. Conditionally on the labels, the $n_k + 1$ nonconformities of the class-k kept variants and of the test variant are exchangeable, so with ties broken at random the rank of the test value is uniform on $\{1, \ldots, n_k + 1\}$ [1, 2]. The test value is at most the j_k-th smallest calibration value with probability $j_k/(n_k + 1)$, which lies in $[1 - \alpha_k,\ 1 - \alpha_k + 1/(n_k + 1))$ for $j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil$; ties only increase the probability, which gives the lower bound in general. If $j_k > n_k$ then $\hat{t}_k = \infty$. Since $k \in C$ exactly when $r_k \le \hat{t}_k$, the claim follows. □

Proposition 2. Conditionally on the labels, the coverage of the threshold is $F_k(\hat{t}_k) = F_k(r^{(k)}_{(j)})$. If F_k is continuous, $F_k(r^{(k)}_{(j)})$ is the j-th order statistic of n_k independent uniforms and has the $\mathrm{Beta}(j, n_k - j + 1)$ distribution, so $\Pr\{F_k(\hat{t}_k) < 1 - \alpha_k\} = F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha_k) \le \delta_{\mathrm{P}}$ by the choice of j_k [3]. If F_k has atoms, write $r = F_k^{-1}(U)$ with U uniform and F_k^{-1} the generalized inverse; then $r^{(k)}_{(j)} = F_k^{-1}(U_{(j)})$ and $F_k(F_k^{-1}(u)) \ge u$, so $F_k(r^{(k)}_{(j)}) \ge U_{(j)}$ and the bound still holds. If no $j \le n_k$ qualifies, $\hat{t}_k = \infty$ and coverage is 1. □

Proposition 3. Fix $\mathcal{A}$. Both ranks in (1) are non-increasing in α (the in-expectation rank because $\lceil (n_k + 1)(1-\alpha) \rceil$ is, the PAC rank because $F_{\mathrm{Beta}(j, n_k - j + 1)}(1-\alpha)$ decreases in α for each j, so the smallest qualifying j cannot increase), so $\hat{t}_k(\alpha)$ is non-increasing in α and the set $\{r_k \le \hat{t}_k(\alpha)\}$ shrinks as α grows. Hence, for any $\hat{\alpha} \in G$ that depends on $\mathcal{A}$ and $\mathcal{B}$ but not on the test variant, $\Pr(k \in C_{\hat{\alpha}} \mid Y = k, \mathcal{A}, \mathcal{B}) \ge \Pr(k \in C_{\max G} \mid Y = k, \mathcal{A})$, and taking expectations over $\mathcal{A}$ and Proposition 1 at $\alpha_k = \max G$ give the first claim. For the PAC rank at level $\delta_{\mathrm{P}}/|G|$, Proposition 2 applied to each $\alpha \in G$ and a union bound give, with probability at least $1 - \delta_{\mathrm{P}}$, coverage at least $1 - \alpha$ for every $\alpha \in G$ simultaneously, in particular for $\hat{\alpha}$. □

Theorem 1. The thresholds $\hat{t}(\alpha)$ depend only on $\mathcal{A}$ and are fixed given $\mathcal{A}$. Conditionally on $\mathcal{A}$ and on the labels of the kept variants of $\mathcal{B}$, the class-1 kept variants of $\mathcal{B}$ are n_1 independent draws from $Q(\cdot \mid Y = 1)$, so the number of them called LoF is Binomial$(n_1, \lambda_1(\alpha))$ with $\lambda_1(\alpha) = \Pr(\text{call LoF} \mid Y = 1)$, and likewise for the other three counts. Each one-sided Clopper–Pearson bound [4] at level $\delta/(2|G|)$ fails with probability at most $\delta/(2|G|)$; the conventions for $x = 0$, $x = n$ and $n = 0$ in Section 3 give the trivial bounds 0 and 1, which never fail. For LR^+ there are $2|G|$ such bounds ($\underline{\lambda}_1$ and $\overline{\lambda}_0$ for each α), so by the union bound all of them hold simultaneously with probability at least $1 - \delta$. On that event $\lambda_1(\alpha) \geq \underline{\lambda}_1(\alpha)$ and $\lambda_0(\alpha) \leq \overline{\lambda}_0(\alpha)$ for every $\alpha \in G$, hence $\mathrm{LR}^+(\hat{t}(\alpha)) \geq L^+(\alpha)$ for every α (with $\mathrm{LR}^+ = \infty$ when $\lambda_0(\alpha) = 0$), including any $\hat{\alpha}^+$ chosen from these data. Since this probability bound holds for every $\mathcal{A}$, it also holds unconditionally. The argument for LR^- is identical with $\overline{\phi}_1$ and $\underline{\phi}_0$. The event does not involve the levels τ, so it covers every evidence level reported. □

Theorem 2. Fix $\mathcal{A}$, so the thresholds and the hypotheses $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \leq \tau$ are fixed. (a) If H_α is true and both endpoints $\underline{\lambda}_1(\alpha)$ and $\overline{\lambda}_0(\alpha)$ at level $\delta'/2$ hold, then $\underline{\lambda}_1 \leq \lambda_1 \leq \tau\lambda_0 \leq \tau\overline{\lambda}_0$, so H_α is not rejected; hence a true H_α is rejected with probability at most δ'. Testing stops at the first non-rejection, so the rejected set is a prefix of the order. Let α^* be the first α in the order for which H_α is true (if there is none, every claim is true). The procedure claims the level at an α where H_α is true only if it rejects every hypothesis up to that α, which includes H_{α^*}; this has probability at most δ'. The selected α is one of the rejected ones, so the claimed level holds with probability at least $1 - \delta'$. (b) With $\delta' = \delta/L$ for each of L levels, a union bound over the levels shows that some claimed level fails with probability at most δ. The functional case is identical with $\overline{\phi}_1/\underline{\phi}_0 < \tau'$. □

Corollary 1. Every quantity bounded above is a probability conditional on the true class: $\Pr(k \in C \mid Y = k)$, the four call rates $\lambda_1, \lambda_0, \phi_1, \phi_0$, and their ratios. Under Assumption 2 these are the same for target variants as for a test variant under Assumption 1, whatever π is, so Propositions 1–3 and Theorems 1–2 transfer. The two call sets are disjoint for any $\alpha^+, \alpha^- \in G$: a LoF call at α^+ requires $q > \hat{t}_0(\alpha^+)$ and $1 - q \leq \hat{t}_1(\alpha^+)$, a functional call at α^- requires $q \leq \hat{t}_0(\alpha^-)$ and $1 - q > \hat{t}_1(\alpha^-)$, and both thresholds are non-increasing in α, so the first conditions contradict each other if $\alpha^- \geq \alpha^+$ and the second if $\alpha^- < \alpha^+$. A functional call on a LoF variant requires $1 \notin C$, so its rate is at most $1 - \Pr(1 \in C \mid Y = 1) \leq \alpha_1$. None of these quantities involves π, so no prior is needed to certify them; this concerns the likelihood ratio with respect to the assay outcome, not evidence about pathogenicity. □

S2 Simulation

Scores were drawn per residue as $s = 2y + u_c + e$, with a residue effect $u_c \sim N(0, 0.5^2)$ shared by the five variants of a residue, unit Gaussian noise, and the class shared within a residue. A Platt calibrator was fitted on 600 variants at LoF share 0.45, conformal thresholds were set on another 600 (one variant per residue), and 3,000 test variants were drawn at LoF shares from 0.05 to 0.60, with 300 replicates per share (Figure S1). Per-class thresholds kept LoF coverage at the target at every share; pooled thresholds did not, and reweighting them [5] to the EM estimate of the share [6] kept marginal coverage near the target while LoF coverage fell further as LoF variants became rarer. Adjusting posteriors to the EM estimate removed the calibration error caused by the shift in this simulation, where the shift is a pure change of prior, unlike in the real data (Section 4.2).

Why marginal thresholds do not control the minority class. A pooled split-conformal threshold $\hat{t}$ is a quantile of the mixture $\pi_c F_1 + (1 - \pi_c)F_0$ of the two class-conditional nonconformity laws, weighted by the calibration share π_c. Its coverage of class 1 is $F_1(\hat{t})$, which is not controlled; when LoF variants are rare and harder to separate, $F_1(\hat{t}) < 1 - \alpha$ while marginal coverage is still near $1 - \alpha$. Reweighting to a known test prior [5] restores marginal coverage at that prior but does not control $F_1(\hat{t})$ either.

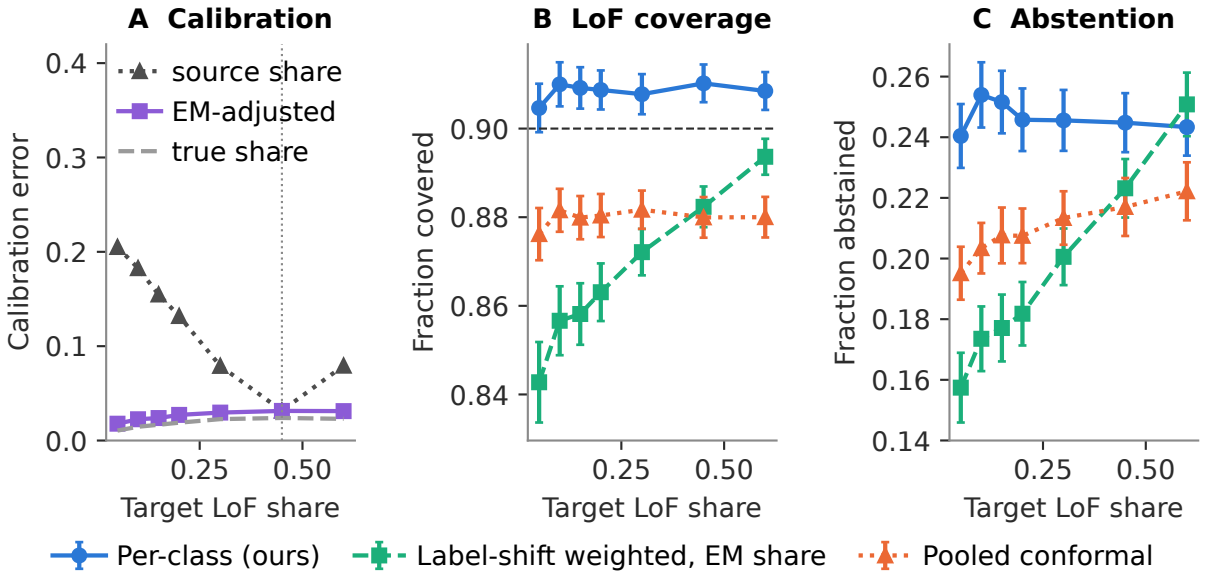


Figure S1: Simulation with residue-clustered scores, calibration at LoF share 0.45 and test share on the horizontal axis (300 replicates; error bars are 95% intervals of the mean). (A) Expected calibration error of Platt scaling at the source share, after EM prior adjustment, and after adjustment to the true share; dotted line: the calibration share 0.45. (B) Coverage of LoF variants. (C) Abstention rate.

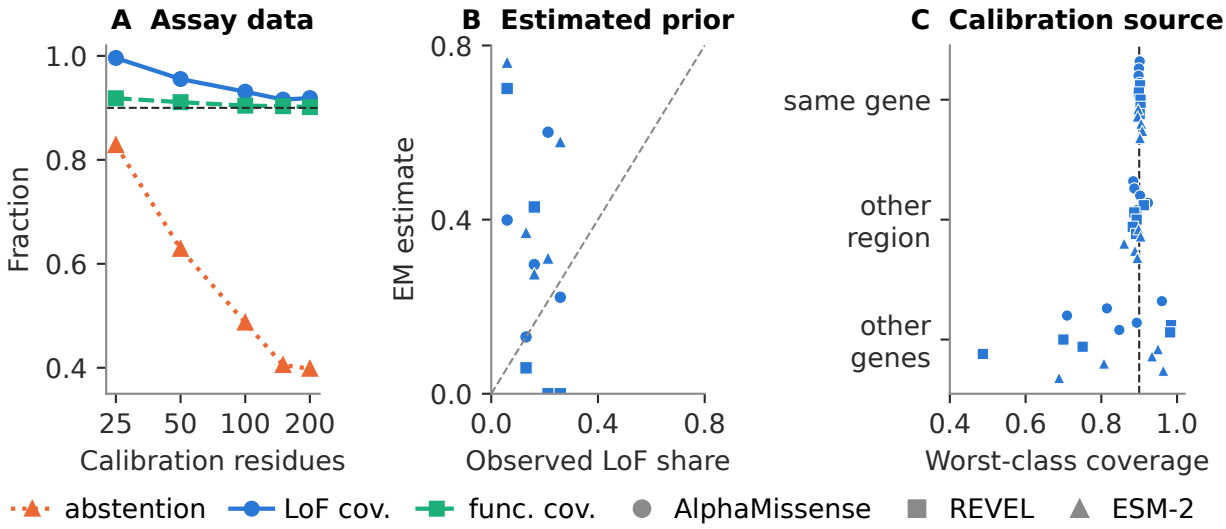


Figure S2: (A) Coverage and abstention against the number of calibration residues drawn from the gene (mean over genes, predictors and 50 draws). (B) EM estimate of the LoF share from unlabelled scores against the observed share; the dashed line is equality. (C) Worst-class coverage when calibrating on random residues of the same gene, on the other three quarters of the protein, or on the other four genes.

S3 Calibration size, region and gene transfer, and per-gene results

Figure S2 shows coverage and abstention against the number of calibration residues, the EM estimates of the LoF share and coverage under region and gene transfer. Table S1 gives per gene and predictor the residue-weighted coverage of each class and the abstention rate of the per-class rule and two baselines; Table S2 gives the estimated LoF shares and calibration errors behind Section 4.2. When the calibration and test halves of a gene share their LoF share, label-shift-weighted conformal prediction [5] with the true share nearly coincides with pooled conformal; with the EM estimate it inherits the estimation error, which collapsed its

LoF coverage in some pairs (worst-class coverage as low as 0.039 at $\alpha = 0.1$).

Table S1: Within-gene random split, $\alpha = 0.1$, mean of 50 splits. Each cell is residue-weighted LoF coverage / functional coverage / abstention rate.

Predictor	Gene	Per-class (ours)	Pooled conformal	Fixed 0.1/0.9
AlphaMissense	BRCA1	0.916/0.901/0.25	0.952/0.889/0.32	0.971/0.927/0.44
AlphaMissense	BRCA2	0.899/0.903/0.36	0.808/0.921/0.17	0.890/0.965/0.37
AlphaMissense	BAP1	0.909/0.898/0.39	0.977/0.880/0.69	0.942/0.687/0.26
AlphaMissense	RAD51C	0.901/0.905/0.17	0.824/0.929/0.10	0.961/0.973/0.37
AlphaMissense	MSH2	0.915/0.901/0.29	0.972/0.897/0.70	0.956/0.799/0.39
REVEL	BRCA1	0.914/0.902/0.34	0.962/0.883/0.48	0.997/0.999/0.98
REVEL	BRCA2	0.912/0.899/0.42	0.841/0.911/0.24	0.947/0.998/0.68
REVEL	BAP1	0.903/0.904/0.42	0.545/0.997/0.15	0.663/1.000/0.24
REVEL	RAD51C	0.909/0.904/0.27	0.625/0.999/0.20	0.680/1.000/0.23
REVEL	MSH2	0.911/0.901/0.37	0.991/0.896/0.79	0.991/0.900/0.80
ESM-2	BRCA1	0.911/0.897/0.37	0.912/0.893/0.34	0.970/0.975/0.64
ESM-2	BRCA2	0.896/0.903/0.36	0.944/0.894/0.51	0.964/0.958/0.71
ESM-2	BAP1	0.910/0.905/0.47	0.878/0.912/0.36	0.903/0.943/0.46
ESM-2	RAD51C	0.915/0.908/0.34	0.973/0.886/0.50	0.987/0.963/0.71
ESM-2	MSH2	0.920/0.902/0.46	1.000/0.895/0.80	0.993/0.823/0.68

Table S2: Prior estimation and calibration on assayed variants outside the ClinVar labelled set. LoF share observed and estimated from unlabelled scores by expectation–maximization (EM) [6] and black-box shift estimation (BBSE) [7]; expected calibration error (10 equal-mass bins) of the ClinVar-Platt posterior, after EM prior adjustment, after adjustment to the true prior, and of Platt scaling fitted on the other four genes' assay labels. AM: AlphaMissense.

		LoF share			Expected calibration error			
Pred.	Gene	obs.	EM	BBSE	ClinVar Platt	+ EM	+ true prior	other genes
AM	BRCA1	0.161	0.297	0.316	0.203	0.135	0.064	0.049
AM	BRCA2	0.130	0.131	0.178	0.132	0.032	0.032	0.041
AM	BAP1	0.213	0.601	0.588	0.341	0.388	0.255	0.119
AM	RAD51C	0.259	0.222	0.252	0.067	0.043	0.037	0.162
AM	MSH2	0.060	0.399	0.395	0.361	0.340	0.120	0.139
REVEL	BRCA1	0.161	0.429	0.336	0.285	0.268	0.058	0.140
REVEL	BRCA2	0.130	0.059	0.149	0.176	0.071	0.025	0.111
REVEL	BAP1	0.213	0.000	0.000	0.092	0.213	0.149	0.138
REVEL	RAD51C	0.259	0.000	0.000	0.143	0.259	0.185	0.187
REVEL	MSH2	0.060	0.701	0.577	0.485	0.642	0.106	0.329
ESM-2	BRCA1	0.161	0.276	0.304	0.212	0.114	0.050	0.044
ESM-2	BRCA2	0.130	0.371	0.371	0.288	0.241	0.075	0.035
ESM-2	BAP1	0.213	0.312	0.327	0.171	0.121	0.086	0.092
ESM-2	RAD51C	0.259	0.579	0.538	0.246	0.320	0.122	0.102
ESM-2	MSH2	0.060	0.762	0.670	0.531	0.702	0.165	0.234

S4 Strata, assay-label sensitivity and ClinVar check

For one within-gene split per gene and predictor ($\alpha = 0.1$, variant-weighted), coverage of LoF and functional variants was 0.966 and 0.929 on the third of variants farthest from the assay's class boundary, 0.929 and

0.912 on the middle third, 0.865 and 0.873 on the nearest third, and 0.881 and 0.896 on variants classified VUS in ClinVar, with abstention rates of 0.334, 0.385, 0.413 and 0.392.

Table S3 gives, for each label rule, the number of labelled variants, their LoF share, the coverage and abstention of the per-class rule, and the oracle check of the certified selection (Section 4.6). This analysis uses its own random splits. Means over 50 random halvings move between runs by up to about 0.02 in coverage and 0.05 in abstention for a high-variance pair (for RAD51C with ESM-2, LoF coverage 0.895 and abstention 0.28 here against 0.915 and 0.34 in Table S1), which explains its differences from Tables S1 and S8.

Table S3: Sensitivity to the assay-label rule. Each gene is relabelled with the authors' classes or with the control rule at the stated posterior cut-off; other genes are unchanged. Per-class: lowest mean per-class coverage over the three predictors (50 random halvings, $\alpha = 0.1$); abstention: range over predictors; bounds held: oracle selections whose certified bound the true likelihood ratio respected (100 draws per predictor); LoF-supp.: share of draws certifying LoF-supporting calls, range over predictors. This is an independent run with its own random splits, so its main-label rows differ from Tables S1 and S8 by Monte Carlo error.

Gene	Labels	n	LoF	Per-class	Abstention	Bounds held	LoF-supp.
BRCA2	controls, 0.80	3,971	0.136	0.894	0.49–0.53	294/295	0.63–0.96
BRCA2	controls, 0.95 (main)	3,435	0.130	0.896	0.36–0.42	423/423	0.90–0.99
BRCA2	controls, 0.99	2,341	0.151	0.897	0.30–0.37	591/593	0.94–1.00
BAP1	authors (main)	4,294	0.213	0.901	0.39–0.47	460/460	0.90–1.00
BAP1	controls, 0.95	4,221	0.100	0.899	0.13–0.19	777/777	1.00
RAD51C	authors (main)	2,225	0.259	0.895	0.18–0.28	558/561	0.84–0.99
RAD51C	controls, 0.95	832	0.538	0.902	0.07–0.19	868/868	0.98–1.00

Table S4 gives the exploratory check of the calls against ClinVar labels (Section 4.4), per gene and pooled. Only BRCA1, BRCA2 and MSH2 have enough labelled residues for informative bounds; across them the LoF-call lower bounds differ by gene, so the pooled values should not be read as a common likelihood ratio. Some ClinVar classifications already use these assays or predictors as evidence, so this check is not independent clinical validation.

Table S5 counts the ClinVar missense VUS of each gene by assay status, and Table S6 gives the certified calls and mixture checks for the 147 that are unassayed but lie at assayed residues (Section 4.5).

Figure S3 shows, for the oracle design of Section 4.4, how often each selection rule claimed each level and how often the claim held.

S5 Implementation

Inputs: ClinVar `variant_summary.txt.gz` of 26 October 2026 (SHA-256 in `data/MANIFEST.txt`); MaveDB score sets `urn:mavedb:00000097-0-2` (BRCA1), `00001279-a-1` (BRCA2), `00000662-0-1` (BAP1), `00000673-0-1` (RAD51C) and `00000050-a-1` (MSH2, positive scores = loss of function); Ensembl VEP REST on GRCh38 (release 116) with the AlphaMissense and REVEL annotations; ESM-2 `esm2_t33_650M_UR50D`. Distance from the class boundary is the absolute difference between a variant's assay score and the midpoint of the two class medians in its gene, divided by the distance between the medians. Expected calibration error uses ten equal-mass bins. AUROC intervals are percentiles of 1,000 bootstrap resamples of residues. The region split divides each gene's assayed residues into four contiguous blocks with equal numbers of distinct positions. ClinVar records whose aggregate germline classification is empty (somatic-only submissions) are treated as unclassified. All random draws use NumPy generators with fixed seeds (2027 plus a fixed offset per experiment). The experiments run on a laptop CPU with Python 3.13, NumPy 2.5, SciPy 1.18, scikit-learn 1.9 and pandas 3.1; ESM-2 scoring used fair-esm 2.0 and PyTorch 2.14 on an Apple-silicon GPU. Exact versions and data checksums are in the repository. The Platt map is checked to be increasing (positive slope) every time it is fitted, and every fit in the experiments passed. Certified bounds use the conventions of Section 3: $U^- = \infty$ when $\phi_0 = 0$, an empirical ratio $x/0$ is ∞ for $x > 0$, and $0/0$ is undefined (never a claim); boundary cases are covered by unit tests (`make test`). The VUS-shift LoF shares are residue-weighted for both the calibration sample (one variant per residue) and the VUS (each residue

Table S4: Calls judged against ClinVar labels (exploratory). Thresholds ($\alpha = 0.1$) are set on each gene's assay labels; one ClinVar variant with $\geq$2 stars is kept per residue. n: kept P/LP and B/LB variants; calls: LoF (or functional) calls among P/LP / among B/LB variants; L^+, U^-: likelihood-ratio bounds from one-sided Clopper–Pearson endpoints at level 0.025 each (simultaneous level 0.95 per ratio). Pooled rows sum the kept counts over genes, each gene with its own thresholds. BAP1 and RAD51C have too few labelled residues to be informative.

Pred.	Gene	n P/LP	n B/LB	LoF calls	L^+	func. calls	U^-
AM	BRCA1	75	170	56/1	19.6	8/153	0.24
AM	BRCA2	60	210	49/8	9.44	6/164	0.29
AM	BAP1	6	8	5/0	0.97	1/8	1.02
AM	RAD51C	8	3	5/1	0.27	2/2	6.90
AM	MSH2	72	55	50/6	2.58	9/46	0.31
AM	pooled	221	446	165/16	11.9	26/373	0.21
REVEL	BRCA1	75	170	48/10	4.94	8/134	0.28
REVEL	BRCA2	60	210	44/9	7.56	11/182	0.37
REVEL	BAP1	6	8	4/0	0.60	2/8	1.23
REVEL	RAD51C	8	3	6/0	0.49	2/3	2.23
REVEL	MSH2	72	55	43/3	3.14	5/38	0.28
REVEL	pooled	221	446	145/22	7.99	28/365	0.23
ESM-2	BRCA1	75	170	47/1	15.7	10/159	0.26
ESM-2	BRCA2	60	210	33/4	8.66	9/184	0.32
ESM-2	BAP1	6	8	5/0	0.97	1/5	2.62
ESM-2	RAD51C	8	3	5/0	0.35	1/1	62.7
ESM-2	MSH2	72	55	44/3	3.23	3/33	0.25
ESM-2	pooled	221	446	134/8	15.4	24/382	0.19

Table S5: ClinVar missense VUS by assay status. Assayed: VUS with an assay score, including intermediate and splice-region variants (Table 1 counts only those with a binary label). Check: share of 20 mixture checks rejected ($p < 0.05$) for the unassayed VUS at other residues, run where they span at least 30 residues.

			Unassayed		Check rejected,
Gene	All	Assayed	at assayed residues	elsewhere	elsewhere (%)
BRCA1	1,909	375	29	1,505	100.0
BRCA2	3,508	1,003	0	2,505	100.0
BAP1	1,189	1,173	3	13	–
RAD51C	596	570	10	16	–
MSH2	1,707	1,597	105	5	–
Total	8,909	4,718	147	4,044	

weighted equally, then its variants).

Comparison rules (Table 3). All rules select, for each level, the grid point with the largest yield whose bound meets the level (functional calls among α at least the LoF choice of the same strength). Uncorrected Clopper–Pearson: endpoints at level $\delta/2$ for every grid point. Katz: the one-sided log interval for a ratio of two binomial proportions [8] at level $\delta/|G|$ per ratio, with 0.5 added to every cell of a ratio that has a zero count. Percentile bootstrap: 2,000 multinomial resamples of the certification residues, the 5th percentile of LR^+ and the 95th of LR^-; corrected: the $100\,\delta/(2|G|)$ percentiles; +0.5 counts: 0.5 added to each of the six cells of every resample. Fixed sequence: levels tested as in Section 3, at δ/L with $L = 3$ (supporting, moderate, strong) for the simultaneous version and at δ for the per-level version.

Table S6: Certified calls on the ClinVar VUS that are unassayed but lie at assayed residues (Algorithm 1, G_4, $\delta = 0.05$; mean number of calls over 20 random 40/60 calibration/certification splits of all assayed residues), and the share of mixture checks of those VUS rejected ($p < 0.05$), run where they span at least 30 residues; where the check was run, calls are averaged over the splits it accepted.

			LoF calls		Functional calls		Check
Gene	VUS	Predictor	supp.	mod.	supp.	mod.	rejected (%)
BRCA1	29	AlphaMissense	7.9	0.2	11.7	0.0	–
		REVEL	7.2	0.0	3.8	0.0	–
		ESM-2	8.0	1.0	5.0	0.0	–
BAP1	3	AlphaMissense	2.9	0.6	0.0	0.0	–
		REVEL	2.7	0.4	0.0	0.0	–
		ESM-2	2.9	0.0	0.0	0.0	–
RAD51C	10	AlphaMissense	2.8	0.8	5.7	1.2	–
		REVEL	3.9	0.5	3.4	0.0	–
		ESM-2	3.4	0.1	4.2	0.0	–
MSH2	105	AlphaMissense	31.8	5.6	60.8	4.8	40.0
		REVEL	20.0	0.0	22.9	0.0	35.0
		ESM-2	19.1	0.8	5.0	0.0	0.0

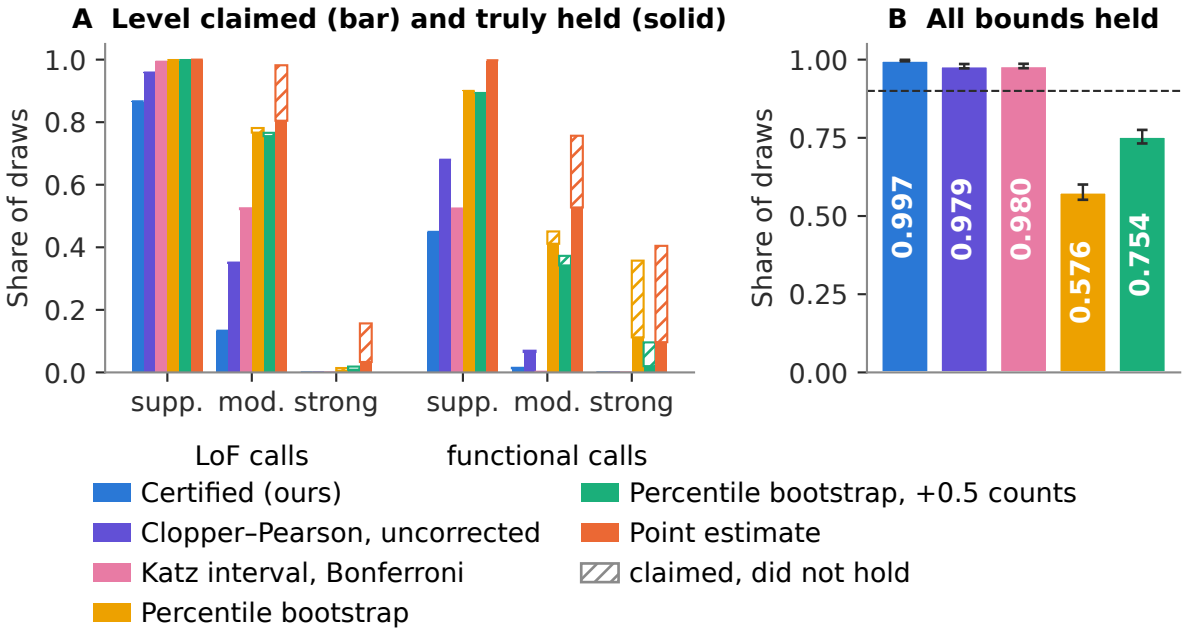


Figure S3: Threshold selection judged against the population likelihood ratio (oracle design, grid G_4, the same 1,500 draws for every rule). (A) Share of draws in which each rule claims a level (full bar) and in which it held (solid). (B) Share of draws in which every claimed bound was respected, with 95% intervals from resampling draws; a point estimate is not a bound and is omitted. Dashed line: the guaranteed 0.90.

Mixture check. The statistic of Section 3 is evaluated on the union of the observed scores, with π on 201 equally spaced values in $[0, 1]$. Each of the 300 bootstrap resamples redraws the two calibration classes with replacement at their observed sizes and a target sample of the observed size from the fitted mixture of the empirical class laws; the p-value is $(1 + \#\{D^* \geq D\})/301$. In a simulation with Gaussian classes the check rejected 4% of 100 target sets that satisfied invariance and 96% of 50 that did not. Calibration (one variant per residue) and target (one per residue) are drawn afresh in every repetition; Table S7 gives the designs.

S6 Experimental designs and all baselines

Table S7 lists the designs of the main text; every split is by residue, and calibration keeps one variant per residue unless stated. Table S8 gives, for every gene and predictor, the per-class rule next to pooled conformal prediction and the published cut-offs.

Table S7: Experimental designs.

Experiment	Residue split	Tests	Reps
Coverage (4.2)	halves: calibrate on one, test on the other	Prop. 1	50
Calibration share (4.2)	halves; calibration half subsampled to LoF share 0.05–0.5	Cor. 1	100
Reliability (4.3)	halves	Props. 1–2	200
VUS (4.3)	halves; non-VUS of one half (40% calibrate, 60% certify); VUS of the other	Cor. 1	50
Population coverage (4.3)	half of the residues drawn with replacement; coverage on all residues	Props. 1–2	200
Weighting (4.2)	halves; one per residue, all variants, residue weights	Prop. 1	50
Oracle (4.4)	30% and 50% of residues drawn with replacement; truth on all residues	Thms. 1–2, Prop. 3	100
Held-out fifth (4.4)	disjoint 30% / 50% / 20%	Thm. 1	100
Mixture check (4.5)	halves; VUS of the other half; quarters; other genes; unassayed VUS	Asm. 2	100/50/25/5/20
Use case (4.5)	all assayed residues, 40% calibrate, 60% certify; unassayed VUS	Thm. 1	20
Regions (4.6)	three contiguous quarters / the fourth	beyond Asm. 1	25/quarter

S7 Coverage and certification figures, and further results

Figure S4 shows per-class coverage by method, per split and on VUS (Sections 4.2 and 4.3); Figure S5 shows coverage under a shift of the calibration LoF share and the certification yields of the Bonferroni and fixed-sequence rules (Sections 4.2 and 4.4).

Further results. In MSH2, with 6% LoF variants, pooled conformal prediction left the functional class marginally below 0.90 (Table S8), so which class a marginal threshold under-covers depends on the gene. Calibrating on all variants of the calibration half instead of one per residue used about five times as many labels and abstained less (30.4% against 35.5%); but in a separate run of 50 halvings that compared the two, worst-class coverage fell below 0.90 in 12 of 15 pairs (lowest 0.885), against 5 (lowest 0.897) for one variant per residue. All-variant calibration weights variants while the test weights residues; weighting each residue equally left 1 pair below 0.90 (lowest 0.898) but abstained more (44.6%), which points to the weighting rather than the dependence, so we keep one variant per residue. Under the calibration-share shift, a separate run balanced at 50% LoF gave worst-class coverage of 0.891–0.925. In the oracle design, claimed levels held in 99.9% of certified claims when the likelihood ratio was weighted by variant rather than by residue; testing each level by fixed sequence at the full δ, which guarantees one level at a time, reached LoF-moderate in 30.5% of draws on G_4; and functional levels tested in increasing α reached functional-supporting in 24.7% of draws against 61.9% in decreasing order. The control label rule left 76.8% of the authors' functional RAD51C variants intermediate, which is why n falls from 2,225 to 832 in Table S3 (`label_rule_comparison` in `paper_numbers.json`). With 150 calibration residues drawn at random, coverage averaged over genes met the target with 40.5% abstention (worst pair 0.890; 82.8% abstention with 25 residues; Figure S2A). Coverage was lower on the third of variants nearest the assay class boundary (0.865 LoF, 0.873 functional; Section S4). Against ClinVar labels (exploratory, not independent of the assays), LoF calls had likelihood-ratio lower bounds of 4.9–19.6 (BRCA1), 7.6–9.4 (BRCA2) and 2.6–3.2 (MSH2) (Table S4).

Table S8: Calls by gene and predictor ($\alpha = 0.1$; calibration and test on disjoint halves of the gene's residues, mean of 50 splits). Per-class rule: residue-weighted coverage of LoF and functional (Fun) variants and abstention rate (Abs); Monte Carlo standard errors of the coverages are 0.002–0.010. Pooled conformal: worst-class coverage and abstention. Published cut-offs (AlphaMissense classes for AM, REVEL PP3/BP4 thresholds for REVEL; none exist for ESM-2): coverage of each class and abstention. Bold: below 0.85.

		Per-class (ours)			Pooled conformal		Published cut-off		
Gene	Predictor	LoF	Fun	Abs	Worst	Abs	LoF	Fun	Abs
BRCA1	AM	0.916	0.901	0.25	0.889	0.32	0.927	**0.773**	0.13
	REVEL	0.914	0.902	0.34	0.883	0.48	1.000	**0.663**	0.59
	ESM-2	0.911	0.897	0.37	0.893	0.34	–	–	–
BRCA2	AM	0.899	0.903	0.36	**0.808**	0.17	**0.810**	0.866	0.11
	REVEL	0.912	0.899	0.42	**0.841**	0.24	0.984	**0.803**	0.57
	ESM-2	0.896	0.903	0.36	0.894	0.51	–	–	–
BAP1	AM	0.909	0.898	0.39	0.880	0.69	0.921	**0.540**	0.07
	REVEL	0.903	0.904	0.42	**0.545**	0.15	**0.799**	0.975	0.29
	ESM-2	0.910	0.905	0.47	0.878	0.36	–	–	–
RAD51C	AM	0.901	0.905	0.17	**0.824**	0.10	0.860	0.902	0.11
	REVEL	0.909	0.904	0.27	**0.625**	0.20	**0.792**	0.979	0.25
	ESM-2	0.915	0.908	0.34	0.886	0.50	–	–	–
MSH2	AM	0.915	0.901	0.29	0.897	0.70	0.951	**0.642**	0.12
	REVEL	0.911	0.901	0.37	0.896	0.79	0.995	**0.445**	0.41
	ESM-2	0.920	0.902	0.46	0.895	0.80	–	–	–

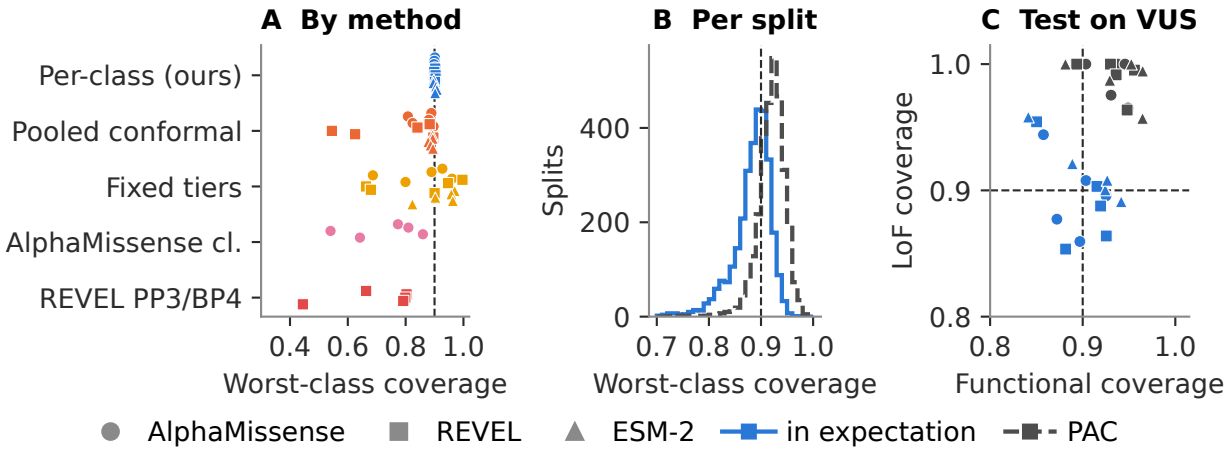


Figure S4: Per-class coverage at $\alpha = 0.1$. Marker shape: predictor (panels A, C); colour and line style: threshold rule (panels B, C). (A) Worst-class coverage for each gene–predictor pair (50 splits); AlphaMissense cl.: the published AlphaMissense classes. (B) Smaller of the two per-class coverages in each test half, pooled over pairs (3,000 splits per rule). (C) Coverage on assayed VUS with thresholds set on non-VUS variants at other residues (50 repetitions).

References

[1] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005. doi: 10.1007/b106715.

[2] Anastasios N. Angelopoulos and Stephen Bates. Conformal Prediction: A Gentle Introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[3] Vladimir Vovk. Conditional validity of inductive conformal predictors. *Machine Learning*, 92(2–3): 349–376, 2013.

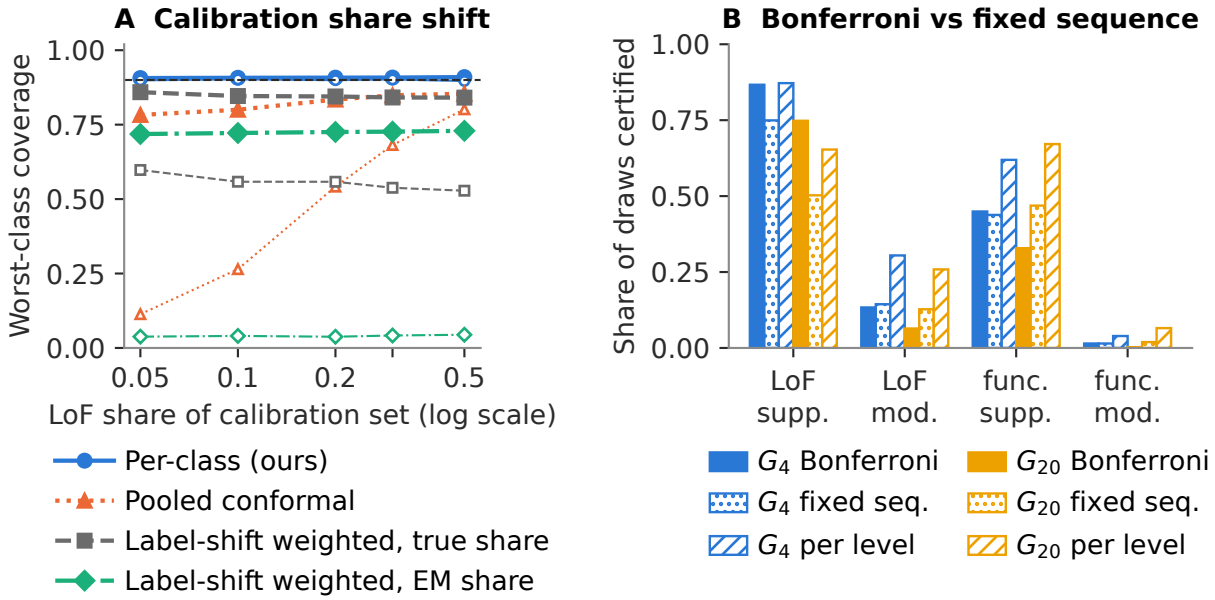


Figure S5: (A) Worst-class coverage when the calibration half is subsampled to a given LoF share (0.05, 0.1, 0.2, 0.3 and 0.5, log scale): mean over 15 pairs (thick) and worst pair (thin, open markers). (B) Share of oracle draws certifying each level with a Bonferroni correction, with fixed-sequence testing simultaneous over levels (fixed seq.), or with fixed-sequence testing of each level at the full δ (per level), on the four-point (G_4) and 20-point (G_{20}) grids.

[4] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.

[5] Aleksandr Podkopaev and Aaditya Ramdas. Distribution-free uncertainty quantification for classification under label shift. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 844–853, 2021.

[6] Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation*, 14(1):21–41, 2002.

[7] Zachary Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3122–3130, 2018.

[8] D. Katz, J. Baptista, S. P. Azen, et al. Obtaining confidence intervals for the risk ratio in cohort studies. *Biometrics*, 34(3):469–474, 1978.